UPPSALA UNIVERSITET

Project : Analysis of Time Series

Time Series Analysis of Electricity Consumption in Île-de-France

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1 Introduction

The study of electricity consumption is a major issue, given the current economic and environmental challenges. Due to the complexity of the signal (lack of stationarity, large volume of data, structural changes caused by external factors, etc.), the study and, in particular, the forecasting of this type of time series remain challenging when using classical models, which are often too general despite still being widely used. However, it is becoming increasingly accessible thanks to advancements in time series analysis, particularly with the rise of LSTM models and machine learning techniques based on artificial intelligence.
One of the primary objectives of our project is to analyze electricity consumption in a French region, specifically Île-de-France, using theoretical time series models such as SARIMA or ARIMA. This initial phase aims to better understand and explain the specific characteristics of the consumption signal.

The next step will be to develop a robust model for predicting weekly electricity consumption in this region.

Our goal is to design a forecasting tool that is as accurate as possible using the available resources. This could offer significant benefits, such as strengthening the reliability of the power grid, supporting policy-making with concrete data, and promoting sustainable energy practices both in France and abroad. Ultimately, such a predictive model would enable us to anticipate fluctuations in energy demand, especially during extreme weather conditions, and help ensure the efficient allocation of electrical resources.

2 Data analysis

The dataset used corresponds to regional electricity consumption curves in France, expressed in megawatts (MW) every 30 minutes, over the period from 2013 to 2024. These data are available as open data on the platform **Open Data Réseaux Énergies (ODRÉ)**.

Launched in 2017, the ODRÉ platform provides a wide range of data covering various domains: production, multi-energy consumption (electricity and gas), storage, mobility, etc. This initiative is part of a transparency approach aimed at citizens, local authorities, and economic stakeholders, to support the development and evaluation of energy policies.

Initially, the dataset includes **2 524 608 values**, along with several variables: temporal data, gross electricity consumption, gas consumption, total consumption, etc., all recorded at 30-minute intervals over multiple years and for each region of France. As part of our study, we chose to focus on the **Île-de-France region**, characterized by a high population density and significant socio-economic heterogeneity. Furthermore, we chose to focus exclusively on the gross **electricity consumption** in this region.

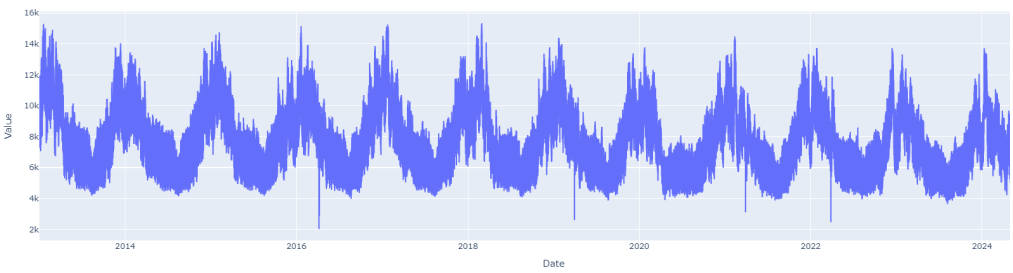

Figure 1: Electricity consumption (MW) in Ile-de-France region for the years 2013 until 2024 (30 min data)

Source: ODRÉ

3 Signal analysis

3.1 Exploratory Data Analysis

As previously observed in the graphs, the overall signal exhibits an oscillating structure due to the strong correlation between electricity consumption and outdoor temperature. The Île-de-France region has a temperate climate, with average winter temperatures ranging from 0 to 10°C and summer temperatures between 15 and 25°C. Consequently, during winter, residents using electric heating tend to consume more energy, whereas in summer, consumption remains lower due to mild temperatures and limited use of air conditioning.

To confirm this hypothesis, the idea was to add a new feature to our dataset: the daily average outdoor temperature in the region. This data is available as open data on the *data.gouv.fr* website, which provides daily average temperatures by region since 2018.

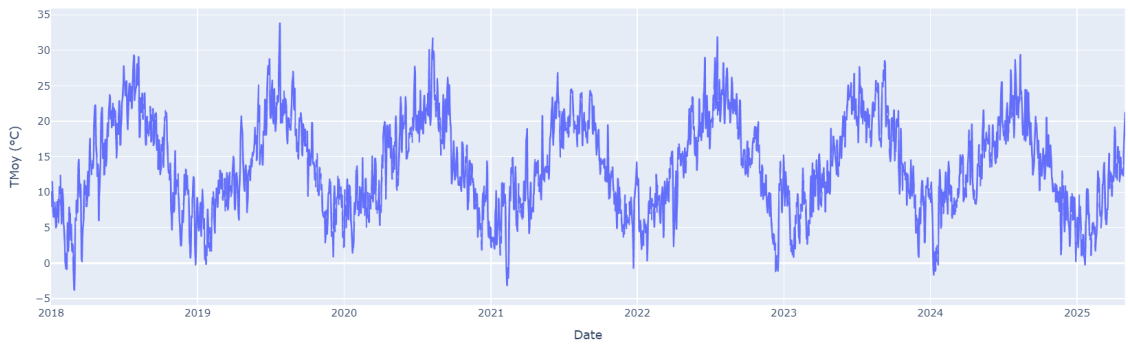

Figure 2: Daily average temperature (°C) in the Île-de-France region

Source: data.gouv.fr

After merging the two dataframes based on the temporal index and studying the correlation between the two variables (Value : electricity consumption and TMoy°C : the average temperature in °C), we obtain the following matrix:

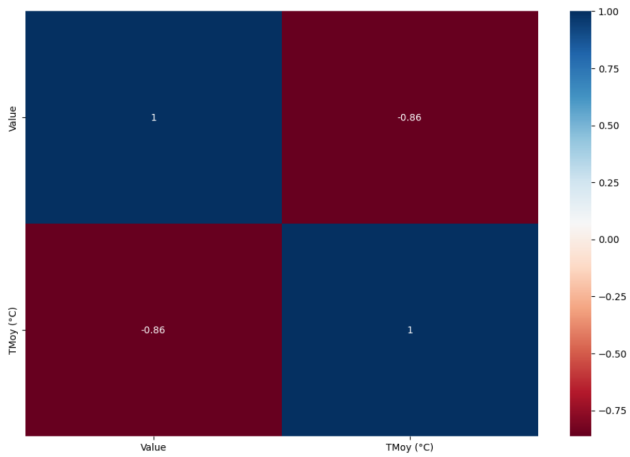

Figure 3: Correlation matrix between the electricity consumption and the average temperature in °C

Source: ODRÉ and data.gouv

Finally, we obtain a strong inverse correlation of -0.86, confirming our point.

3.2 Analysis of the First Signal: Short Time Frame and Daily Data

Due to the large size of the dataset, we needed to adapt our approach. Initially, we chose to retain only specific years 2022 and 2023 based on both deliberate and practical considerations. These years represent a recent period post-COVID, characterized by consistent consumption patterns and complete data without missing values. Additionally, despite European conflicts such as the war in Ukraine, electricity consumption in Île-de-France appeared largely unaffected, likely because of the significant share of nuclear energy in the French energy mix. We then aggregated the data on a daily basis to calculate average daily consumption and because of the large variance, we applied a logarithmic transformation to the data.

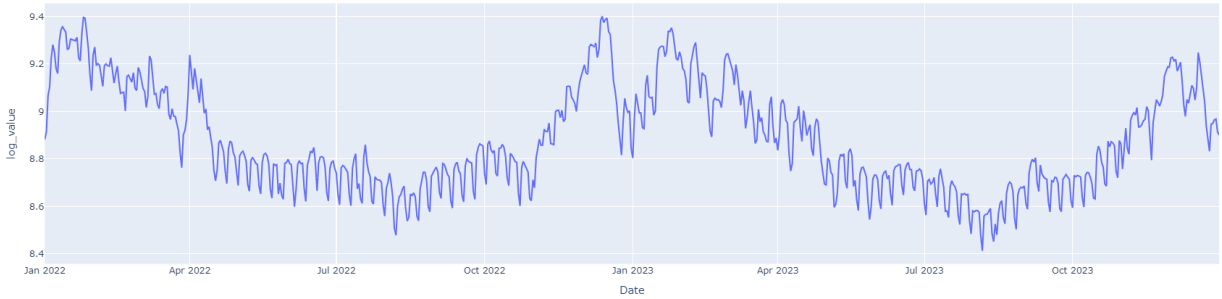

Figure 4: Electricity consumption (MW) in Ile-de-France region for the years 2022 until 2023 (1 Day data)

Source: ODRÉ

The signal [4] therefore appears to be non-stationary, due to multiple seasonal trends and recurring patterns in weekly consumption cycles, given the daily time step.

To verify this, we begin by analyzing the ACF (slowly decaying) and PACF (cutting off) plots :

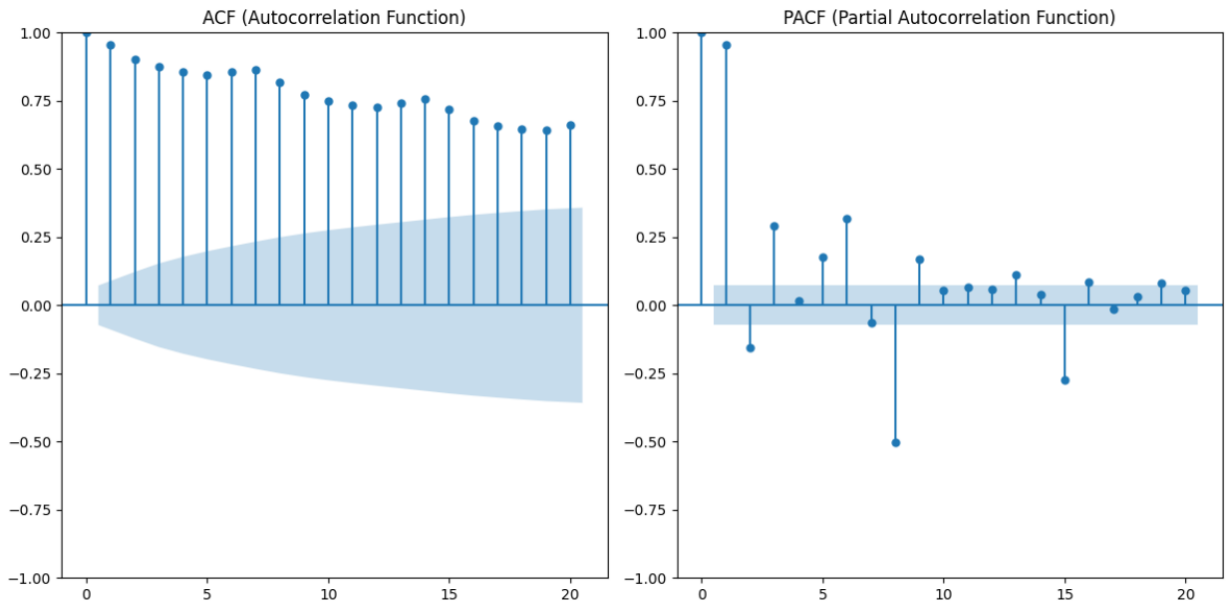

Figure 5: ACF and PCF plots

Source: ODRÉ

Furthermore, the **ADF test (Dickey-Fuller)** confirms that the signal is not stationary.

3.2.1 Differencing

Due to the non-stationarity and seasonal patterns of the original signal, one of the first steps was to difference the series.

Initially, due to the daily nature of the data, a seasonal differencing with a period of s=7, corresponding to a weekly pattern, was applied.

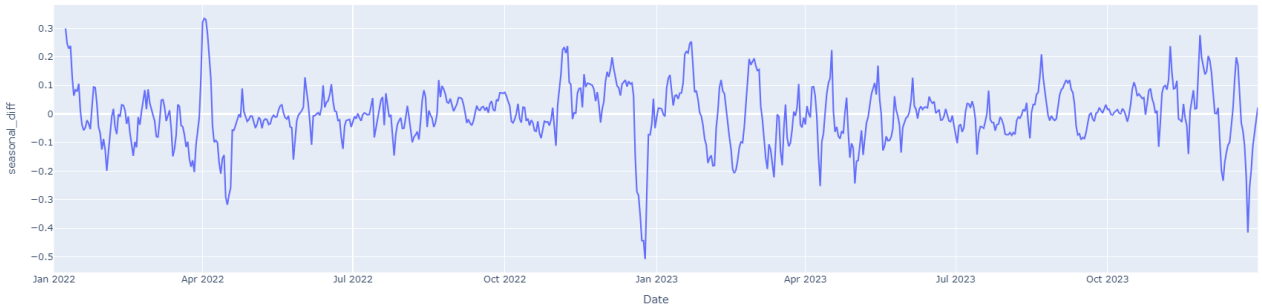

Figure 6: First seasonal (s=7) Differcing signal

Source: ODRÉ

As well as a regular differencing.

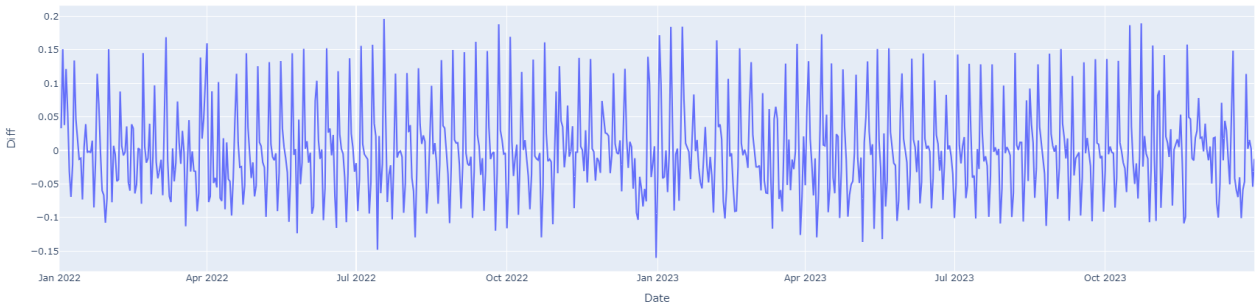

Figure 7: First regular Differcing signal

Source: ODRÉ

And finally, both were applied.

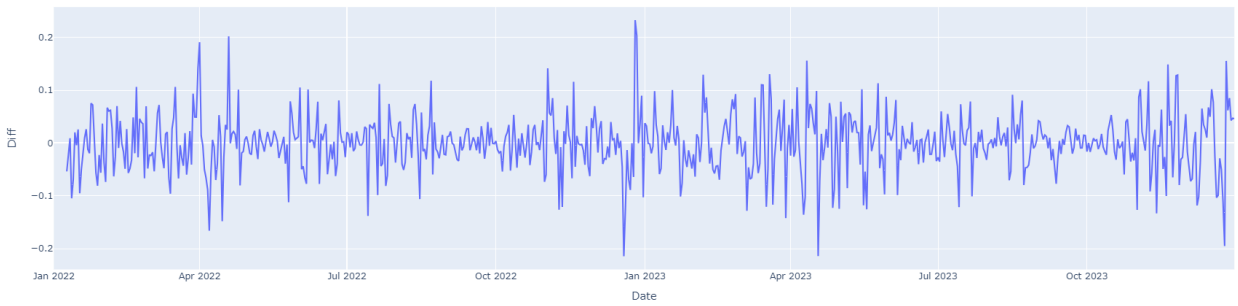

Figure 8: First seasonal and regular Differcing signal

Source: ODRÉ

This time, the ADF test confirms the stationarity of the series (p-value : 8.489e-17 < 0.05).

3.2.2 Model building

To determine the parameters of the future SARIMA(p,d,q)(P,D,Q)s model, one should start by analyzing the autocorrelation function (ACF) and partial autocorrelation function (PACF) of the differenced signals.

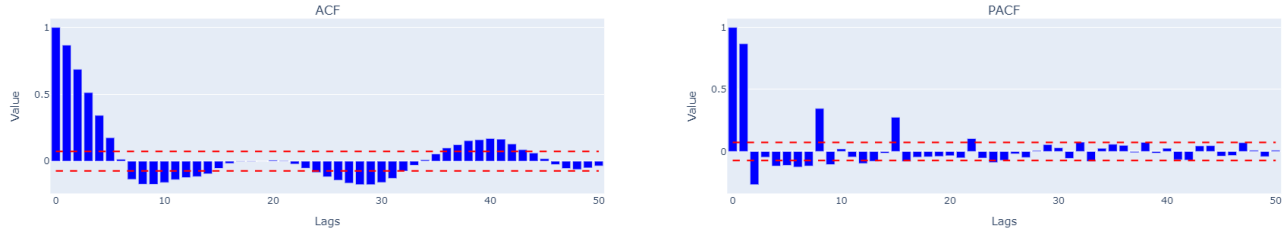

Figure 9: ACF and PACF : First seasonal (s=7) differcing signal

Source: ODRÉ

In the case of the ACF, significant spikes are observed at multiples of 7 (7, 14, 21, . . .), which is typical of weekly seasonality. The signal therefore retains a strong seasonal component. The PACF shows a rapid decline after the first lags, with a notable spike at lag 1 and some spikes at lags that are multiples of 7. This indicates that the partial dependence is well addressed in the short term, but traces of seasonality remain.

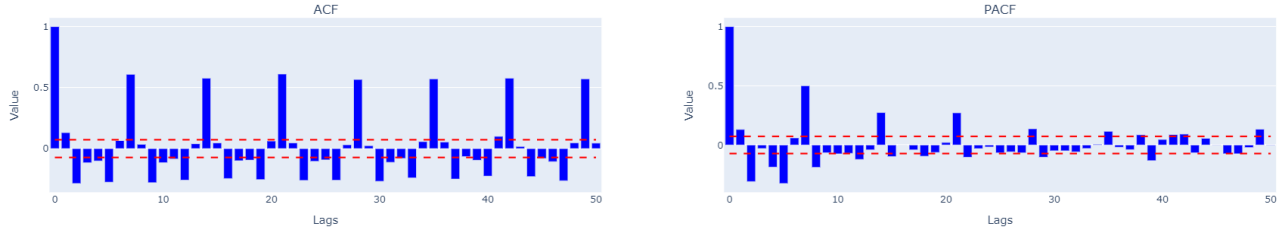

Figure 10: ACF and PACF : First regular differcing signal

Source: ODRÉ

The ACF shows a strong alternation between positive and negative values, which can indicate a process close to an AR(1) model. Seasonality is still present but less pronounced. Meanwhile, the PACF shows a strong spike at lag 1, then declines rapidly.

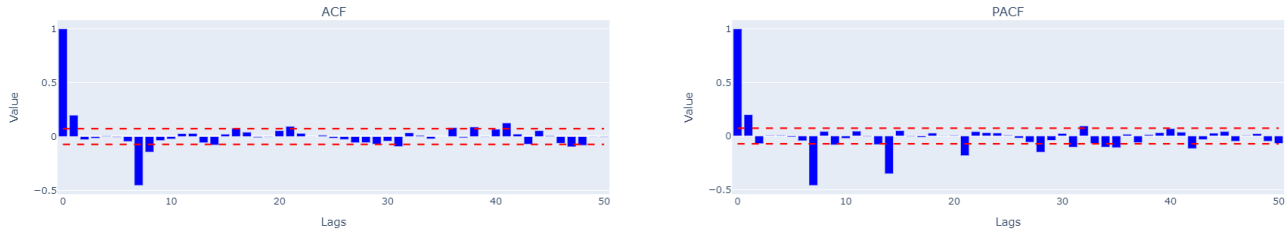

Figure 11: ACF and PACF : First seasonal and regular differcing signal

Source: ODRÉ

In the case of the ACF, we are now closer to white noise, with values mostly within the confidence bands, suggesting an absence of significant residual autocorrelation. The same observation applies to the PACF, with no significant spikes. The signal now appears to be stationary.

3.2.3 Model Selection

After further testing to find an appropriate model, the final selected one that produced the best and most accurate results was an $SARIMA(0,1,1)(0,1,0)_7$. This model yielded an AIC of -2168.419, which was the lowest among all the models tested, indicating the best fit to the data.

```
                                     SARIMAX Results
==========================================================================================
Dep. Variable:                          log_value   No. Observations:                  730
Model:             SARIMAX(0, 1, 1)x(0, 1, [], 7)   Log Likelihood                1086.209
Date:                            Sat, 17 May 2025   AIC                          -2168.419
Time:                                    15:17:09   BIC                          -2159.260
Sample:                                01-01-2022   HQIC                         -2164.883
                                     - 12-31-2023
Covariance Type:                              opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ma.L1          0.1544      0.028      5.474      0.000       0.099       0.210
sigma2         0.0029      0.000     27.030      0.000       0.003       0.003
===================================================================================
Ljung-Box (L1) (Q):                   0.02   Jarque-Bera (JB):               137.52
Prob(Q):                              0.88   Prob(JB):                         0.00
Heteroskedasticity (H):               0.99   Skew:                             0.03
Prob(H) (two-sided):                  0.93   Kurtosis:                         5.14
===================================================================================
```

Figure 12: Parameter estimates

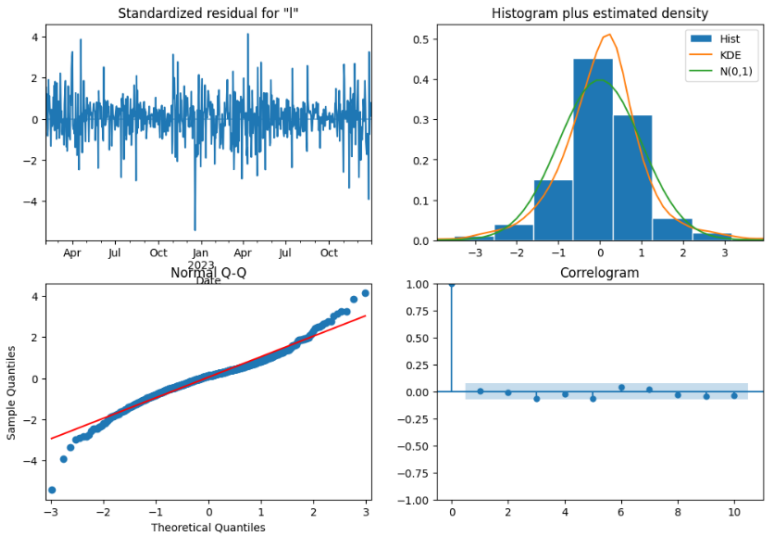

Figure 13: Residual plots for daily model selection

Source: ODRÉ

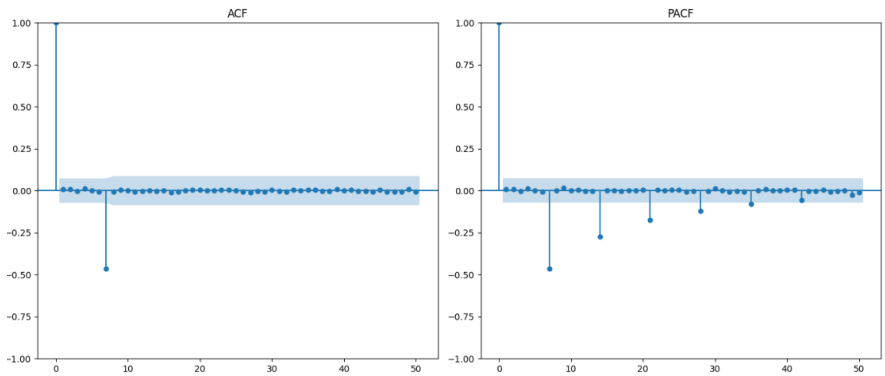

Figure 14: Residual plots of ACF and PACF

Source: ODRÉ

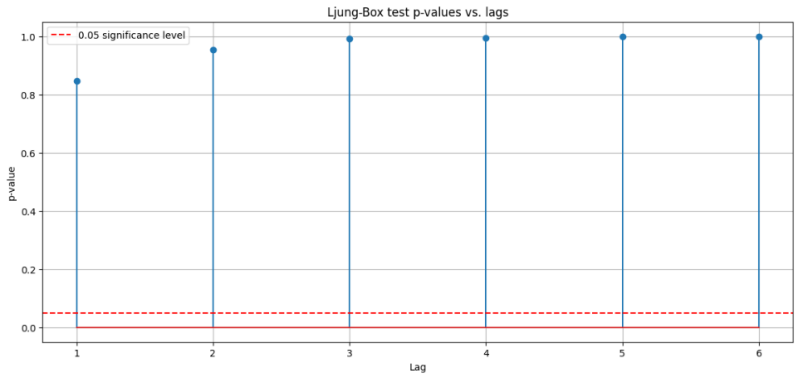

Figure 15: Ljung-box test

Source: ODRÉ

After analyzing the residuals, the model appears to perform well: the Q-Q plot aligns closely with the theoretical distribution despite a few outliers, and no discernible pattern is observed in the residuals. Furthermore, the Ljung-Box test indicates no significant short-term autocorrelation in the residuals (at least up to lag 6). This finding is supported by the ACF and PACF plots, which show a rapid decline in autocorrelations and partial autocorrelations.

3.3 Analysis of the Second Signal: Long Time Frame and Weekly Data

3.3.1 Data processing

In a second step, in order to make the most of the information provided by the dataset, we also used the full period from 2013 to 2022, this time with a weekly aggregation.

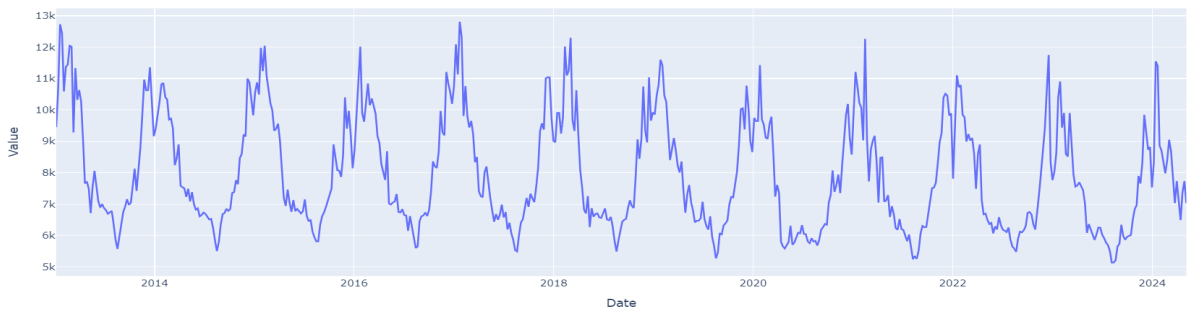

Figure 16: Electricity consumption (MW) in Ile-de-France region for the years 2013 until 2022 (weekly aggregation)

Source: ODRÉ

As shown in the graph [16], electricity consumption follows a clear pattern, likely influenced by seasonal factors such as temperature and public holidays. Aggregating the data on a weekly basis helps smooth out short-term fluctuations, providing a clearer view of long-term trends.

Before analyzing the weekly patterns, we converted the electricity consumption values from megawatts (MW) to gigawatts (GW) for consistency and readability. Additionally, because of the large variance, we applied a logarithmic transformation to the data. This helped reduce the effect of extreme values and made the underlying patterns more visually and statistically clear.

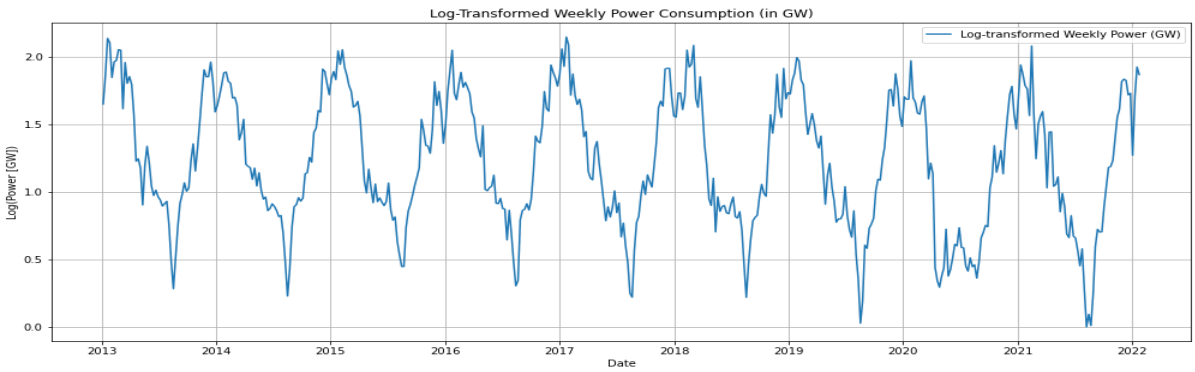

Figure 17: Log-Transformed Weekly electricity Consumption (in GW)

Source: ODRÉ

3.3.2 Stationarity Analysis

Given that the signal appears non-stationary due to periodic seasonal fluctuations, we performed ACF and PACF analysis on the weekly series:

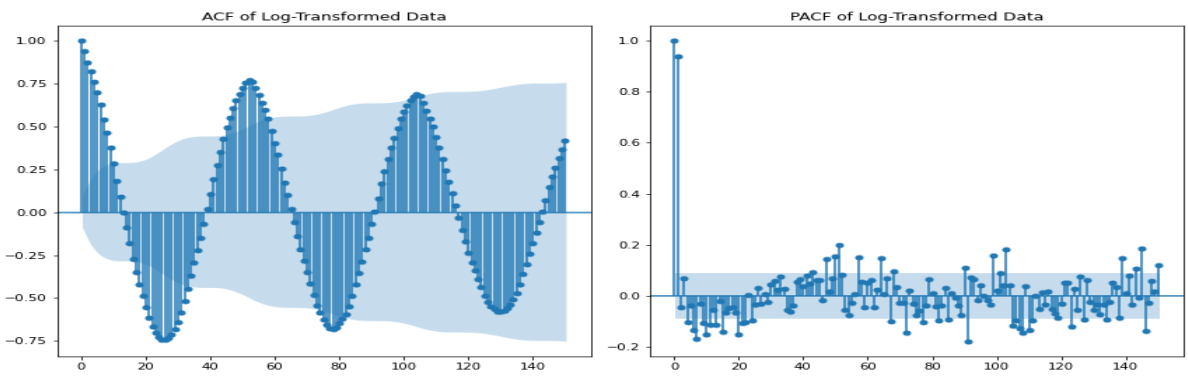

Figure 18: ACF and PACF for Log-Transformed weekly electricity consumption (in GW)

Source: ODRÉ

As indicated by the ACF and PACF plots, the series exhibits significant seasonal behavior. To address this, we applied seasonal differencing to remove the seasonal patterns. Additionally, to ensure stationarity of the time series, we performed one nonseasonal difference. After these transformations, we conducted the Augmented Dickey-Fuller (ADF) test. The test yielded a statistic of -12.517 and a p-value of 2.59×10^{-13} which is significantly less than the common significance level of 0.05. This confirms that the time series is now stationary.

3.3.3 Model building

For the weekly data, we aim to fit an SARIMA model with seasonal adjustments. The ACF and PACF analysis for the series will guide the selection of the model parameters.

- From ACF plot:
  - Sinusoidal decay (alternating positive/negative values) means a seasonal AR component likely needed.
  - Peaks at multiples of ≈ 52 weeks so yearly seasonality (s=52).
- From PACF plot:
  - Small coefficients after lag 1 so no higher-order AR terms needed .

– 1 significant spikes.

Based on the ACF and PACF plot analysis, we initially considered a SARIMA$(1,0,0)(1,0,0)_{52}$ model for the log-transformed weekly electricity consumption data. However, since both one seasonal and one nonseasonal differencing were necessary to achieve stationarity, the model was updated to SARIMA$(1,1,0)(1,1,0)_{52}$ model. We then compared this model with several alternatives. Among these, the SARIMA$(1,1,1)(1,1,0)_{52}$ proved to be the best fitting overall. This was supported by its lower Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) values compared to other models. Further diagnostics using the Ljung-Box test indicated that the SARIMA$(1,1,0)(1,1,0)_{52}$ exhibited significant autocorrelation at higher lags; specifically, its p-values dipped below 0.05 after only lag 10. In contrast, the SARIMA$(1,1,1)(1,1,0)_{52}$ model maintained Ljung-Box p-values well above 0.05 even after lag 10, indicating a better fit with residuals consistent with white noise. Therefore, we decided to select the SARIMA$(1,1,1)(1,1,0)_{52}$ model.

3.3.4 Model Selection

The SARIMA$(1,1,1)(1,1,0)_{52}$ model provided a better fit according to the evaluation of model selection criteria such as AIC, BIC, the residual diagnostics and the Ljung-box test as seen in the Figure below (Figures 19,20 21 and 22).

```
                                     SARIMAX Results
==========================================================================================
Dep. Variable:                                Value   No. Observations:                  501
Model:             SARIMAX(1, 1, 1)x(1, 1, [], 52)   Log Likelihood                 155.428
Date:                              Fri, 16 May 2025   AIC                           -302.856
Time:                                      23:43:56   BIC                           -286.940
Sample:                                  01-06-2013   HQIC                          -296.550
                                       - 08-07-2022
Covariance Type:                                opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          0.4849      0.046     10.453      0.000       0.394       0.576
ma.L1         -0.9562      0.018    -54.249      0.000      -0.991      -0.922
ar.S.L52      -0.5576      0.030    -18.302      0.000      -0.617      -0.498
sigma2         0.0266      0.002     17.447      0.000       0.024       0.030
===================================================================================
Ljung-Box (L1) (Q):                   0.00   Jarque-Bera (JB):                33.96
Prob(Q):                              0.98   Prob(JB):                         0.00
Heteroskedasticity (H):               2.36   Skew:                            -0.08
Prob(H) (two-sided):                  0.00   Kurtosis:                         4.43
===================================================================================
```

Figure 19: Summary of the $SARIMAX(1,1,1)(1,1,0)_{52}$ Results

Source: ODRÉ

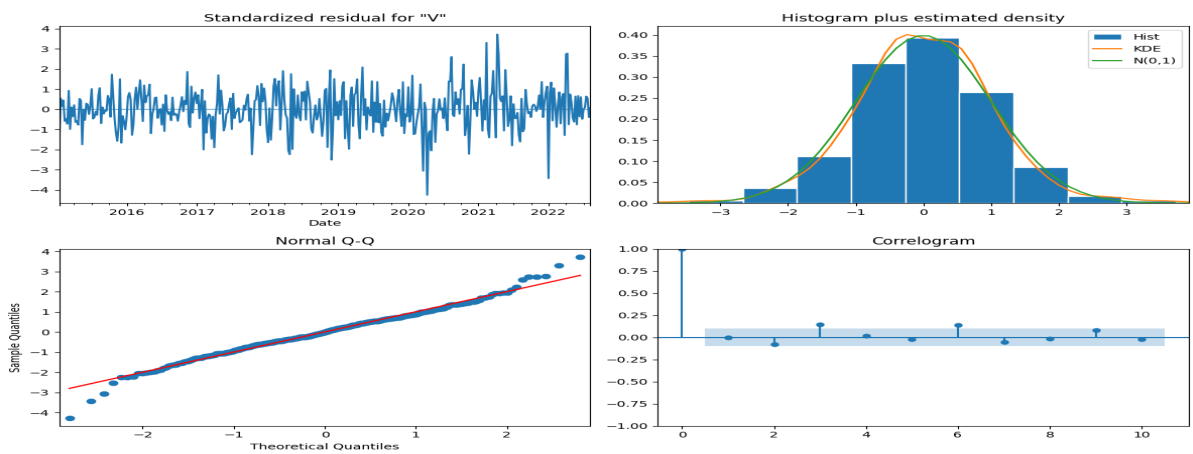

Figure 20: Residual plots

Source: ODRÉ

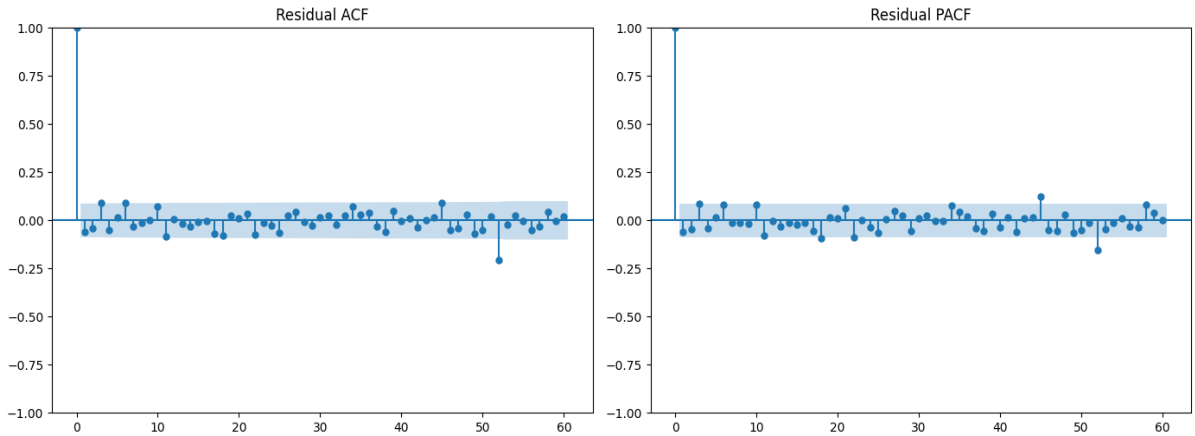

Figure 21: Residual plots of ACF and PACF

Source: ODRÉ

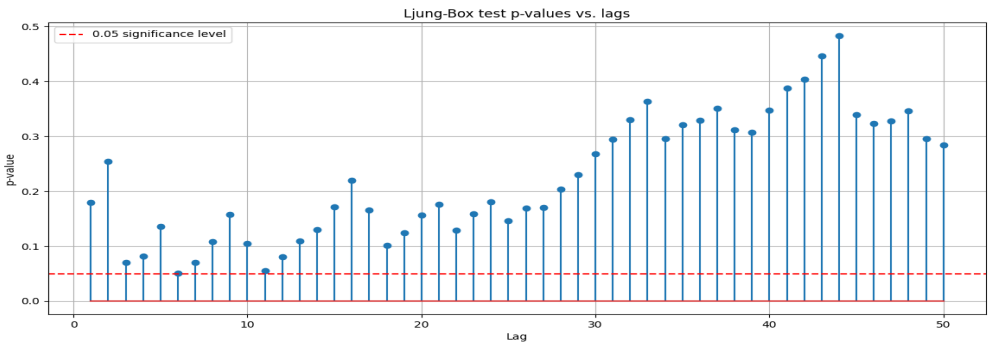

Figure 22: Ljung-box test

Source: ODRÉ

Despite the fact that residuals exhibit mild non-normality, which is expected in electricity consumption data, the Q-Q plot shows only a few outliers and does not invalidate the model. The model remains valid as:

- No significant autocorrelation exists in the residuals, as indicated by resudial plots of ACF and PACF and confirmed by the Ljung-box p-values plot.(Figure22).
- No significant partial autocorrelations.
- Residual plots(Figure21) of ACF and PACF also confirms that the model have has successfully captured all seasonal patterns and trend components.
- The histogram of the residuals closely resembles a normal distribution.

Given the satisfactory performance of this model, we decided to proceed with SARIMA $(1,1,1)(1,1,0)_{52}$ model for forecasting.

4 Forecasting

In this section, we focused on the second signal, using long time frame and weekly data. The selected $SARIMA(1,1,1)(1,1,0)_{52}$ model was employed to generate two years ahead forecasts (covering 2022–2024) for weekly average electricity consumption in GW in Île-de-France. To evaluate the model's forecasting performance, we used first 80% of the available data from 2014 to 2024 as the training set, while the remaining 20% (2022-2024) was used for validation.

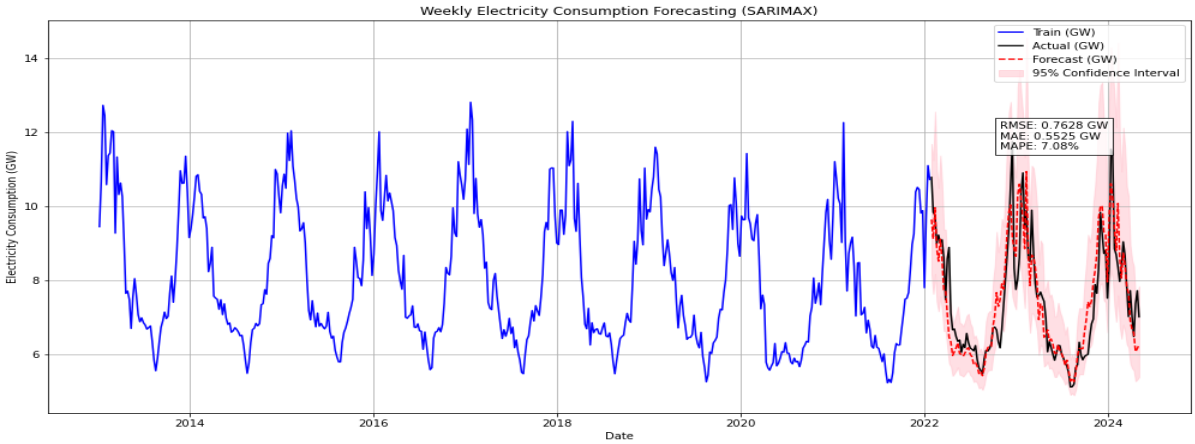

Figure 23: Forecasting of weekly average electricity consumption in (GW)

Source: ODRÉ

As shown in Figure 23, the SARIMA model captured the underlying seasonal and trend components effectively, producing forecasts that closely follow the observed data. The Mean Absolute Percentage Error (MAPE) of 7.08%, indicated a high accuracy. Additional metrics, such as RMSE (0.76 GW) and MAE (0.55 GW), further confirmed the model's reliability. The 95% confidence interval (CI) provided a reasonable range of uncertainty around the predictions.

5 Discussion and Conclusion

In this project, we aimed to address our main objective: to study a time series modeling electricity consumption and to develop a reliable forecasting tool. To do this, we defined several study parameters, including the choice of a single region in France, the analysis period, and the data aggregation method.

We analyzed electricity consumption in Île-de-France using SARIMA models applied to data aggregated both daily (2022–2023) and weekly (2013–2024). The ultimately selected model,$SARIMA(1,1,1)(1,1,0)_{52}$, was used to generate two-year forecasts (covering 2022–2024) of average weekly electricity consumption in gigawatts (GW). Total electricity consumption over a given period can be estimated by multiplying the average consumption by the number of hours.

The weekly model achieved a Mean Absolute Percentage Error (MAPE) of 7.06% during forecasting, with residuals showing no significant autocorrelation, confirming its robustness. It effectively captured both the trend and seasonality of the signal.

Future work could involve integrating more advanced machine learning algorithms that incorporate correlated external variables such as temperature, sunlight, public holidays, and vacation periods to improve forecasting accuracy.

References

1. Larsson, R. (2025). *Lecture notes for Time Series Analysis*. Uppsala University.
2. Shumway, R. H., & Stoffer, D. S. (2025). *Time Series Analysis and Its Applications: With R Examples* (5th ed.). Springer.

Appendix

Appendix - Code

In this appendix, we have Python code used to perform data analysis and plot Figures for the project. All the required python code are order below.

```python
# Packages

import numpy as np
import pandas as pd
import matplotlib as mpl
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.tsa.stattools import adfuller

from statsmodels.tsa.stattools import acf, pacf
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.stattools import acf, pacf




# Data Preprocessing

df = pd.read_csv('consommation-quotidienne-brute-regionale.csv', sep=';')



# Clean column names
df.columns = df.columns.str.strip()

# Select and rename relevant columns
df = df[['Date - Heure', 'Consommation brute    lectricit    (MW) - RTE']]
df = df.rename(columns={
    'Date - Heure': 'Date',
    'Consommation brute    lectricit    (MW) - RTE': 'Value'
})

# Convert date and clean up
df['Date'] = pd.to_datetime(df['Date'])
df.set_index('Date', inplace=True)
df.index = pd.to_datetime(df.index, utc=True)
df.index = df.index.tz_localize(None)
df = df.sort_index()
df = df.dropna()


# Preview
#print(df.head())

# To plot time series
# plt.figure(figsize=(12, 6))
# plt.plot(df.index, df["Value"])
# plt.title("Electricity consumption (MW) in Ile-de-France region for the years 2013 until 2024 (30 min data)")
# plt.xlabel("Date")
# plt.ylabel("Value")
# plt.grid(True)
# plt.tight_layout()
# plt.show()


```

```python
# Create new columns from the datetime index
df['Year'] = df.index.year
df['Month'] = df.index.month
df['Day'] = df.index.day
df['Hour'] = df.index.hour

# Define time of day categories
def categorize_time_of_day(hour):
    if 6 <= hour < 12:
        return 'Morning'
    elif 12 <= hour < 18:
        return 'Afternoon'
    elif 18 <= hour < 24:
        return 'Evening'
    else:
        return 'Night'

df['TimeOfDay'] = df['Hour'].apply(categorize_time_of_day)

# Add weekday (0 = Monday, 6 = Sunday)
df['Weekday'] = df.index.weekday

# Indicate if it's a weekend
df['IsWeekend'] = df['Weekday'].apply(lambda x: int(x >= 5))

# Define seasons
def categorize_season(month):
    if month in (12, 1, 2):
        return 'Winter'
    elif month in (3, 4, 5):
        return 'Spring'
    elif month in (6, 7, 8):
        return 'Summer'
    else:
        return 'Autumn'

df['Season'] = df['Month'].apply(categorize_season)

# Ensure proper data types
df['Year'] = df['Year'].astype(int)
df['Weekday'] = df['Weekday'].astype(int)
df['IsWeekend'] = df['IsWeekend'].astype(int)

# Convert categorical variables to numerical values using One-Hot Encoding
df = pd.get_dummies(df, columns=['TimeOfDay', 'Season'])

# Convert the columns resulting from One-Hot Encoding to integer values
df = df.astype(int)

df.head()



# sampling to weekly frequency using mean (weekly average)
df_weekly = df['Value'].resample('W').mean()
```

```
# Take the first 80% of the weekly data based on the number of rows
n_weeks_80 = int(len(df_weekly) * 0.8)
df_weekly_80 = df_weekly.iloc[:n_weeks_80]

# # Plot the first 80% of weekly data
# plt.figure(figsize=(12, 6))
# plt.plot(df_weekly_80.index, df_weekly_80.values, color='green', linewidth=1.2)
# plt.title("Weekly Electricity Consumption (MW) in  le -de-France (2014  2022  )")
# plt.xlabel("Week")
# plt.ylabel("Average Weekly Consumption (MW)")
# plt.grid(True)
# plt.tight_layout()
# #plt.savefig('weeklydata.png')
# plt.show()





# Convert MW to GW
df_weekly_gw = df_weekly_80 / 1000  # Convert MW to GW

# # Plot the data in GW
# plt.figure(figsize=(12, 6))
# plt.plot(df_weekly_gw.index, df_weekly_gw.values, color='green', linewidth=1.2)
# plt.title("Weekly Electricity Consumption (GW) in  le -de-France (2014  2022  )")
# plt.xlabel("Week")
# plt.ylabel("Average Weekly Consumption (GW)")
# plt.grid(True)
# plt.tight_layout()
# #plt.savefig('weeklydata_GW.png')
# plt.show()





# Log-transform the weekly data
df_weekly_gw_log = np.log(df_weekly_gw - df_weekly_gw.min() + 1)

# #Plotting the log-transformed data
# plt.figure(figsize=(14, 6))
# plt.plot(df_weekly_gw_log.index, df_weekly_gw_log.values, label='Log-transformed Weekly Power (GW)')
# plt.title('Log-Transformed Weekly Power Consumption (in GW)')
# plt.xlabel('Date')
# plt.ylabel('Log(Power [GW])')
# plt.legend()
# plt.grid(True)
# plt.tight_layout()
# #plt.savefig('logtransformed.png')
# plt.show()
```

```python
# # Plot the ACF and PACF for the log-transformed data
# plt.figure(figsize=(12, 6))
# plt.subplot(121)
# plot_acf(df_weekly_gw_log, lags=150, ax=plt.gca(), title="ACF of Log-Transformed Data")
# plt.subplot(122)
# plot_pacf(df_weekly_gw_log, lags=150, ax=plt.gca(), title="PACF of Log-Transformed Data")
# plt.tight_layout()
# #plt.savefig('acfpcflog.png')
# plt.show()



#  non seasonal and seasonal differencing
df_diff =  df_weekly_gw_log.diff().dropna()
df_diff_seasonal = df_diff.diff(52).dropna()



#  Perform ADF test
adf_result = adfuller(df_weekly_gw_log)

# Print ADF test result
print(f"ADF Statistic: {adf_result[0]}")
print(f"p-value: {adf_result[1]}")
print(f"Critical Values: {adf_result[4]}")



# plotting ACF and PACF after differencing
fig, ax = plt.subplots(2, 1, figsize=(12,8))

plot_acf(df_diff_seasonal, lags=120, ax=ax[0])
ax[0].set_title('ACF after Non-seasonal and Seasonal Differencing')

plot_pacf(df_diff_seasonal, lags=120, ax=ax[1])
ax[1].set_title('PACF after Non-seasonal and Seasonal Differencing')

plt.tight_layout()
plt.show()



# # Calculate the variance of the log-transformed data
variance_log = np.var(df_weekly_gw_log)
print(f"Variance of the log-transformed data: {variance_log:.4f}")



from statsmodels.tsa.statespace.sarimax import SARIMAX

```

```python



# Fit SARIMAX model on  data
model = SARIMAX(df_weekly_gw_log,
                    order=(1, 1, 0),
                    seasonal_order=(1, 1, 0, 52),
                    enforce_stationarity=False,
                    enforce_invertibility=False)

results = model.fit(disp=False)

# Print model summary
print(results.summary())

#plots the residual plots
results.plot_diagnostics(figsize=(12, 8))
plt.tight_layout()
plt.show()


# Get residuals from the fitted model
residuals = results.resid

# Plot ACF of residuals (Autocorrelation)
plot_acf(residuals, lags=52, title="Residual ACF")

# Plot PACF of residuals (Partial Autocorrelation)
plot_pacf(residuals, lags=52, title="Residual PACF")


plt.tight_layout()
plt.show()




# For forecasting
from sklearn.metrics import mean_squared_error
import matplotlib.dates as mdates



# 1. Full weekly data:
df_weekly = df['Value'].resample('W').mean()

# 2. Split once (80% train, 20% test)
n_weeks_80 = int(len(df_weekly) * 0.8)
train_raw = df_weekly.iloc[:n_weeks_80]
test_raw = df_weekly.iloc[n_weeks_80:]

# 3. Convert to GW
train_gw = train_raw / 1000
test_gw = test_raw / 1000

# 4. Log-transform both consistently (using the min of train to avoid data
    leakage)
min_train_gw = train_gw.min()
train_log = np.log(train_gw - min_train_gw + 1)
test_log = np.log(test_gw - min_train_gw + 1)

# 5. Fit SARIMAX on train_log
```

```python
model_log = SARIMAX(train_log,
                    order=(1, 1, 0),
                    seasonal_order=(1, 1, 0, 52),
                    enforce_stationarity=False,
                    enforce_invertibility=False)
results_log = model_log.fit(disp=False)

# 6. Forecast steps = length of test set
forecast = results_log.get_forecast(steps=len(test_log))
predicted_mean = forecast.predicted_mean
conf_int = forecast.conf_int()

# 7. Calculate MSE on test vs predicted_mean
from sklearn.metrics import mean_squared_error
mse = mean_squared_error(test_log, predicted_mean)
print(f"MSE on test set: {mse:.4f}")

# 8. Plot
plt.figure(figsize=(14, 7))
plt.plot(train_log.index, train_log, label='Train (log GW)')
plt.plot(test_log.index, test_log, label='Test (log GW)')
plt.plot(predicted_mean.index, predicted_mean, label='Forecast')
plt.fill_between(conf_int.index, conf_int.iloc[:, 0], conf_int.iloc[:, 1], color
    ='pink', alpha=0.3)
plt.legend()
plt.grid(True)
plt.show()
```

Listing 1: Python Code for data analysis